



Final Task(s) – 04 Weightage

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Syllabus:

Operating Systems Lab

Syllabus Code: CL-2006

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Instructor:

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Q1)

```

muhammadabubakar@muhammadabubakar-Zenbook-UX3404VA-Q420VA:~/OSLAB2026/finaltasks$ gcc q1.c -o q1 -pthread
muhammadabubakar@muhammadabubakar-Zenbook-UX3404VA-Q420VA:~/OSLAB2026/finaltasks$ ./q1
1 3 5 12 18 22 45 67 89 99
muhammadabubakar@muhammadabubakar-Zenbook-UX3404VA-Q420VA:~/OSLAB2026/finaltasks$ S

```

Q2)

```

muhammadabubakar@muhammadabubakar-Zenbook-UX3404VA-Q420VA:~/OSLAB2026/finaltasks$ gcc q2.c -o q2 -pthread
muhammadabubakar@muhammadabubakar-Zenbook-UX3404VA-Q420VA:~/OSLAB2026/finaltasks$ ./q2
enter values for first 3x3 matrix:
46 78 232 12 44 90 0 26 1
enter values for second 3x3 matrix:
1 2 3 4 56 78 8 9 2
result matrix is:
2214 6548 6686
908 3298 3648
112 1465 2030
muhammadabubakar@muhammadabubakar-Zenbook-UX3404VA-Q420VA:~/OSLAB2026/finaltasks$ █

```

Q3)

```

muhammadabubakar@muhammadabubakar-Zenbook-UX3404VA-Q420VA:~/OSLAB2026/finaltasks$ gedit q3.c
muhammadabubakar@muhammadabubakar-Zenbook-UX3404VA-Q420VA:~/OSLAB2026/finaltasks$ gcc q3.c -o q3
muhammadabubakar@muhammadabubakar-Zenbook-UX3404VA-Q420VA:~/OSLAB2026/finaltasks$ ./q3
Running
Running
^C
SIGINT caught, terminating,,...
muhammadabubakar@muhammadabubakar-Zenbook-UX3404VA-Q420VA:~/OSLAB2026/finaltasks$

```

Q4)

```

muhammadabubakar@muhammadabubakar-Zenbook-UX3404VA-Q420VA:~/OSLAB2026/finaltasks$ gedit q4.c
muhammadabubakar@muhammadabubakar-Zenbook-UX3404VA-Q420VA:~/OSLAB2026/finaltasks$ gcc q4.c -o q4
muhammadabubakar@muhammadabubakar-Zenbook-UX3404VA-Q420VA:~/OSLAB2026/finaltasks$ ./q4
10...
9...
8...
^CTimer is pawsed. press ctrl+z to resume.
^Z
Timer resumed.
7...
6...
5...
4...
^CTimer is pawsed. press ctrl+z to resume.
^Z
Timer resumed.
3...
2...
1...
^CTimer is pawsed. press ctrl+z to resume.
Done!
muhammadabubakar@muhammadabubakar-Zenbook-UX3404VA-Q420VA:~/OSLAB2026/finaltasks$

```

Q5)

Main Terminal

```

muhammadabubakar@muhammadabubakar-Zenbook-UX3404VA-Q420VA: ~/OSLAB2026/finaltasks$ ./q5
process id is: 6161
Working...
Working...
Working...
Working...
Working...
Working...
Paused by SIGUSR1
Paused by SIGUSR1
Paused by SIGUSR1
Working...
Working...
Working...
Working...
Paused by SIGUSR1
Working...
Working...
Working...
^Z
[1]+  Stopped                  ./q5
muhammadabubakar@muhammadabubakar-Zenbook-UX3404VA-Q420VA: ~/OSLAB2026/finaltasks$

```

Secondary Terminal

```

muhammadabubakar@muhammadabubakar-Zenbook-UX3404VA-Q420VA: ~/OSLAB2026/finaltasks$ kill -SIGUSR1 6161
muhammadabubakar@muhammadabubakar-Zenbook-UX3404VA-Q420VA: ~/OSLAB2026/finaltasks$ kill -SIGUSR1 6161
muhammadabubakar@muhammadabubakar-Zenbook-UX3404VA-Q420VA: ~/OSLAB2026/finaltasks$ kill -SIGUSR1 6161
muhammadabubakar@muhammadabubakar-Zenbook-UX3404VA-Q420VA: ~/OSLAB2026/finaltasks$ kill -SIGUSR1 6161
muhammadabubakar@muhammadabubakar-Zenbook-UX3404VA-Q420VA: ~/OSLAB2026/finaltasks$

```

Q6)

```

muhammadabubakar@muhammadabubakar-Zenbook-UX3404VA-Q420VA: ~/OSLAB2026/finaltasks$ gcc q6.c -o q6 -pthread
muhammadabubakar@muhammadabubakar-Zenbook-UX3404VA-Q420VA: ~/OSLAB2026/finaltasks$ ./q6
philosopher 0 is thinking
philosopher 4 is thinking
philosopher 2 is thinking
philosopher 1 is thinking
philosopher 3 is thinking
philosopher 0 is eating
philosopher 2 is eating
philosopher 0 is thinking
philosopher 2 is thinking
philosopher 4 is eating
philosopher 1 is eating
philosopher 4 is thinking
philosopher 1 is thinking
philosopher 0 is eating
philosopher 3 is eating
philosopher 3 is thinking
philosopher 4 is eating
philosopher 2 is eating
philosopher 0 is thinking
philosopher 4 is thinking
philosopher 3 is eating
philosopher 1 is eating
philosopher 2 is thinking
philosopher 1 is thinking
philosopher 3 is thinking
philosopher 0 is eating
philosopher 2 is eating
^C
muhammadabubakar@muhammadabubakar-Zenbook-UX3404VA-Q420VA: ~/OSLAB2026/finaltasks$

```

Q7)

```
muhammadabubakar@muhammadabubakar-Zenbook-UX3404VA-Q420VA:~/OSLAB2026/finaltasks$ gcc q7.c -o q7 -pthread
muhammadabubakar@muhammadabubakar-Zenbook-UX3404VA-Q420VA:~/OSLAB2026/finaltasks$ ./q7
Vehicle-1 parked - occupied slots: 1 / 4
Vehicle-3 parked - occupied slots: 2 / 4
Vehicle-4 parked - occupied slots: 3 / 4
Vehicle-2 parked - occupied slots: 4 / 4
Vehicle-4 departed - occupied slots: 3 / 4
Vehicle-2 departed - occupied slots: 2 / 4
Vehicle-6 parked - occupied slots: 3 / 4
Vehicle-5 parked - occupied slots: 4 / 4
Vehicle-1 departed - occupied slots: 3 / 4
Vehicle-7 parked - occupied slots: 4 / 4
Vehicle-5 departed - occupied slots: 3 / 4
Vehicle-3 departed - occupied slots: 2 / 4
Vehicle-6 departed - occupied slots: 1 / 4
Vehicle-7 departed - occupied slots: 0 / 4
total revenue collected: 644
muhammadabubakar@muhammadabubakar-Zenbook-UX3404VA-Q420VA:~/OSLAB2026/finaltasks$
```

Q8)

```
muhammadabubakar@muhammadabubakar-Zenbook-UX3404VA-Q420VA:~/OSLAB2026/finaltasks$ gedit q8.c
muhammadabubakar@muhammadabubakar-Zenbook-UX3404VA-Q420VA:~/OSLAB2026/finaltasks$ gcc q8.c -o q8
muhammadabubakar@muhammadabubakar-Zenbook-UX3404VA-Q420VA:~/OSLAB2026/finaltasks$ ./q8
Detector: temperature reading = 65°C
Temperature normal - monitoring continues.
Detector: temperature reading = 78°C
Temperature normal - monitoring continues.
Detector: temperature reading = 85°C
FIRE ALERT! Activating sprinkler system in zone 3!
Detector: temperature reading = 92°C
FIRE ALERT! Activating sprinkler system in zone 4!
Detector: temperature reading = 70°C
Temperature normal - monitoring continues.
Detector: temperature reading = 81°C
FIRE ALERT! Activating sprinkler system in zone 6!
Total suppression events: 3
muhammadabubakar@muhammadabubakar-Zenbook-UX3404VA-Q420VA:~/OSLAB2026/finaltasks$
```

Q9)

Main Terminal

```
muhammadabubakar@muhammadabubakar-Zenbook-UX3404VA-Q420VA:~/OSLAB2026/finaltasks$ gcc q9.c -o q9
muhammadabubakar@muhammadabubakar-Zenbook-UX3404VA-Q420VA:~/OSLAB2026/finaltasks$ ./q9
irrigation system started. process id: 3076
Irrigation system running...
Irrigation system running...
Irrigation system running...
Irrigation system running...
Irrigation system running...
Irrigation system running...
Irrigation system running...

Alert! Soil moisture critically low - activating irrigation!
Irrigation system running...
Irrigation system running...
Irrigation system running...

Alert! Soil moisture critically low - activating irrigation!
Irrigation system running...
Irrigation system running...
Irrigation system running...
Irrigation system running...

Irrigation system powered off safely...
muhammadabubakar@muhammadabubakar-Zenbook-UX3404VA-Q420VA:~/OSLAB2026/finaltasks$
```

Secondary Terminal

```
muhammadabubakar@muhammadabubakar-Zenbook-UX3404VA-Q420VA:~/OSLAB2026/finaltasks$ kill -SIGUSR1 3076
muhammadabubakar@muhammadabubakar-Zenbook-UX3404VA-Q420VA:~/OSLAB2026/finaltasks$ kill -SIGUSR1 3076
muhammadabubakar@muhammadabubakar-Zenbook-UX3404VA-Q420VA:~/OSLAB2026/finaltasks$ kill -SIGTERM 3076
muhammadabubakar@muhammadabubakar-Zenbook-UX3404VA-Q420VA:~/OSLAB2026/finaltasks$
```